



Guangdong Technion

Israel Institute of Technology

广东以色列理工学院

GTIIT

CALCULUS 1 - M
WINTER SEMESTER - 2024

LECTURE 15 - DECEMBER 2ND

2ND PART

- Integrals using Symmetries: Odd and Even Functions
- Indefinite Integrals and the Net Change Theorem (Section 5.4)
- Areas enclosed by Curves. (Section 6.1)

APPLICATIONS OF THE FUNDAMENTAL THEOREM OF CALCULUS

The Fundamental Theorem of Calculus

Let f be continuous on $[a, b]$, then the integral function

① $F(x) = \int_a^x f(t) dt$ is an antiderivative of f on (a, b) .

Equivalently, $\frac{d}{dx} \int_a^x f(t) dt = f(x)$ on (a, b) .

② If F is any antiderivative, that is, $F' = f$,

$$\int_a^b f(x) dx = F(b) - F(a).$$

③ $\int_a^x F'(t) dt = F(x) - F(a)$ on (a, b)

④ $F(x) = \int f(x) dx$ means $F'(x) = f(x)$ on an interval

Differentiation and Integration are Inverse Processes

$$\frac{d}{dx} \int_a^x f(t) dt = f(x) \text{ on } (a, b)$$

$$\int_a^x F'(t) dt = F(x) - F(a) \text{ on } (a, b)$$

The Fundamental Theorem of Calculus

Example 1. Find the 2nd derivative

$$\frac{d^2}{dx^2} \int_0^x \left[\int_0^{\cosh(u)} \sqrt{2+t^6} dt \right] du$$

Solution. Apply the theorem twice: start finding the first derivative

$$\frac{d^2}{dx^2} \int_0^x \underbrace{\left[\int_0^{g(u)} f(t) dt \right]}_{=H(u)} du = \frac{d}{dx} \left[\frac{d}{dx} \int_0^x H(u) du \right] = \frac{d}{dx} H(x)$$

$$= \frac{d}{dx} \left[\int_0^{\cosh(x)} \sqrt{2+t^6} dt \right] = \sqrt{2 + (\cosh(x))^6} \sinh(x) \checkmark$$

The Fundamental Theorem of Calculus for a composition

$$\frac{d}{dx} \left[\int_a^{g(x)} f(t) dt \right] = f(g(x)) \cdot g'(x)$$

Corollary:

$$\frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x)$$

Proof. Splitting by any value $x = c$ on the domain of f , the given integral may be rewritten as a sum of two terms, for which we can apply the theorem:

$$\int_{g(x)}^{h(x)} f(t) dt = \int_{g(x)}^c f(t) dt + \int_c^{h(x)} f(t) dt = - \int_c^{g(x)} f(t) dt + \int_c^{h(x)} f(t) dt$$

Then, the derivative takes the following form

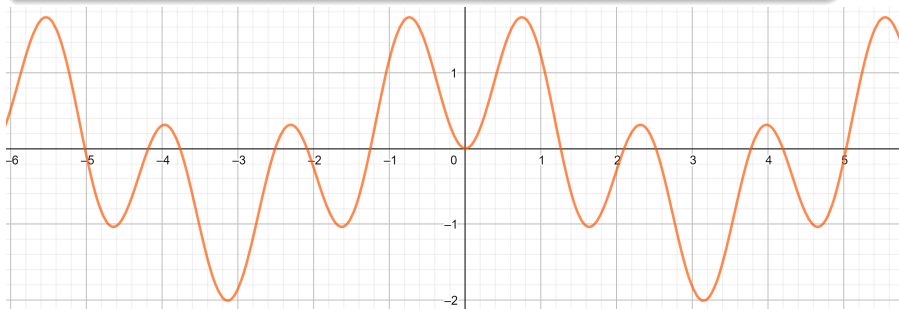
$$\frac{d}{dx} \left[\int_{g(x)}^{h(x)} f(t) dt \right] = f(h(x)) \cdot h'(x) - f(g(x)) \cdot g'(x) \quad \checkmark$$

1. Integral of Symmetric Functions: even functions

Suppose that f is continuous on $[-a, a]$.

If f is **even**, that is, $f(-x) = f(x)$, then

$$\int_{-a}^a f(x) dx = 2 \int_0^a f(x) dx$$

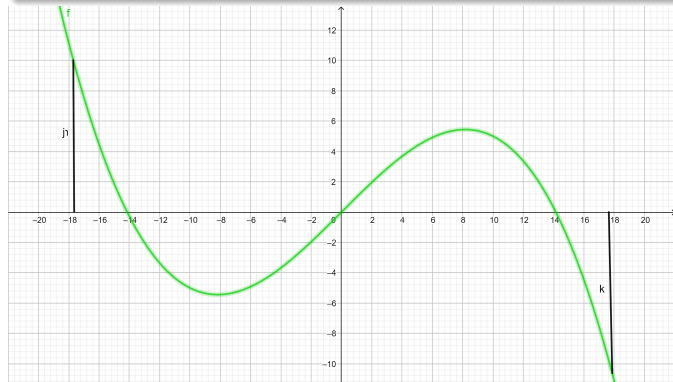


2. Integral of Symmetric Functions: odd functions

Suppose that f is continuous on $[-a, a]$.

If f is **odd**, that is, $f(-x) = -f(x)$, then

$$\int_{-a}^a f(x) dx = 0$$



Proofs.

Use the substitution theorem for definite integrals

For 1) and 2), split the integral in two:

$$\int_{-a}^a f(x) dx = \int_{-a}^0 f(x) dx + \int_0^a f(x) dx = - \int_0^{-a} f(x) dx + \int_0^a f(x) dx$$

On the 1st integral, make the substitution $u(x) = -x$
 $\Rightarrow du = -dx$ and $u(-a) = a$, so the **1st term (negative)**:

$$- \int_0^{-a} f(x) dx = - \int_0^a f(-u) (-du) = \int_0^a f(-u) du$$

Therefore, the given integral becomes

$$\boxed{\int_{-a}^a f(x) dx} = - \int_0^{-a} f(x) dx + \int_0^a f(x) dx = \int_0^a f(-u) du + \int_0^a f(x) dx$$

- If f is even, $f(-u) = f(u)$, so

$$\int_{-a}^a f(x) dx = \int_0^a f(u) du + \int_0^a f(x) dx = 2 \int_0^a f(x) dx$$

- If f is odd, $f(-u) = -f(u)$, so

$$\int_{-a}^a f(x) dx = \int_0^a (-f(u)) du + \int_0^a f(x) dx$$

$$= -\int_0^a f(x) dx + \int_0^a f(x) dx = 0 \quad \checkmark$$

Ex. 1. Solve

$$\int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} [5x^4 + \cos(x)] dx$$

Solution. $f(x) = 5x^4 + \cos(x)$ is even, that is:

$$f(-x) = 5(-x)^4 + \cos(-x) = 5x^4 + \cos(x) = f(x)$$

$$\Rightarrow \int_{-\frac{3\pi}{4}}^{\frac{3\pi}{4}} [5x^4 + \cos(x)] dx = 2 \int_0^{\frac{3\pi}{4}} f(x) dx$$

$$= [\sin(x) + x^5] \Big|_0^{\frac{3\pi}{4}} = \sin\left(\frac{3\pi}{4}\right) + \left(\frac{3\pi}{4}\right)^5 - [\sin(0) + 0^5]$$

$$= \frac{\sqrt{2}}{2} + \left(\frac{3\pi}{4}\right)^5 \checkmark$$

Ex. 2. Solve the integral analyzing the odd and even terms

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\frac{\tan(x)}{x^6 + \cos(x)} + \sqrt[3]{\pi^2 - |x|} \right] dx$$

Solution. Separate in **even** and **odd** terms:

- $g(x) = \frac{\tan(x)}{x^6 + \cos(x)}$ is **odd**:

$$g(-x) = \frac{\tan(-x)}{(-x)^6 + \cos(-x)} = \frac{-\tan(x)}{x^6 + \cos(x)} = -g(x) \checkmark$$

- $f(x) = \sqrt[3]{\pi^2 - |x|}$ is **even**:

$$f(-x) = \sqrt[3]{\pi^2 - |-x|} = \sqrt[3]{\pi^2 - |x|} = f(x) \checkmark$$

The integral is the sum of the integrals of the odd and even terms

$$\int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\frac{\tan(x)}{x^6 + \cos(x)} + \sqrt[3]{\pi^2 - |x|} \right] dx$$

$$\begin{aligned} \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} [g(x) + f(x)] dx &= \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} g(x) dx + \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} f(x) dx \\ &= 0 + 2 \int_0^{\frac{\pi}{4}} f(x) dx \end{aligned}$$

Give a name to the antiderivative $F = \int f$, so the previous is equal to

$$\begin{aligned} & \int_{-\frac{\pi}{4}}^{\frac{\pi}{4}} \left[\frac{\tan(x)}{x^6 + \cos(x)} + \sqrt[3]{\pi^2 - |x|} \right] dx \\ &= 0 + 2 \left[F\left(\frac{\pi}{4}\right) - F(0) \right] \\ &= -2 \frac{3}{4} \left(\pi^2 - x \right)^{4/3} \Big|_0^{\frac{\pi}{4}} = -\frac{3}{2} \left(\pi^2 - \frac{\pi}{4} \right)^{4/3} + \frac{3}{2} \left(\pi^2 \right)^{4/3} \\ &= \frac{3}{2} \left[- \left(\pi^2 - \frac{\pi}{4} \right)^{4/3} + \pi^{8/3} \right] \checkmark \end{aligned}$$

Net Change Theorem

The integral of a rate of change is called the net change:

$$\int_a^b F'(x)dx = F(b) - F(a)$$

Example 3. a) Find the **displacement** of an object during the time $[0, 2]$, if it moves along a straight line with velocity $v(t) = 2 - 2t$.

b) Under the same conditions, find the traveled distance.

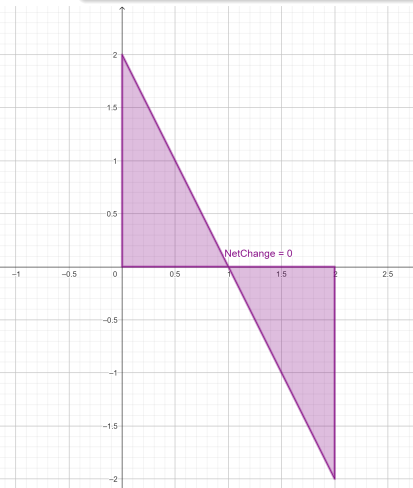
Solution. • The **net change of position is the displacement**, of the particle during the time period from $t = a$ to $t = b$:

$$\text{Net change of position} = s(2) - s(0) = \int_0^2 s'(t)dt = \int_0^2 v(t)dt$$

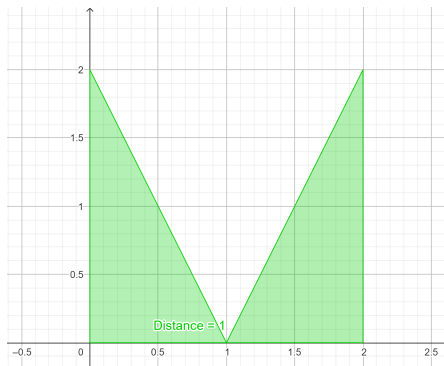
• The **Total Distance traveled** during the time period $[a, b]$:

$$\text{Total Distance Traveled} = \int_a^b |v(t)|dt$$

Net Change Theorem



$$\text{Displacement} = \int v(t) dt$$



$$\text{The Distance} = \int |v(t)| dt$$

Net Change Theorem

Solution of the Example 3.

- The **net change of position is the displacement**, of the particle during the time period from $t = 0$ to $t = 2$:

$$\begin{aligned}\text{Net change of position} &= s(2) - s(0) = \int_0^2 s'(t) dt = \int_0^2 v(t) dt \\ &= \int_0^2 (2 - 2t) dt = \left(2t - t^2 \right) \Big|_{t=0}^{t=2} = 2(\textcolor{red}{2}) - \textcolor{red}{2}^2 - 0 = 0\end{aligned}$$

This means that the object started and finished at the **same place**.

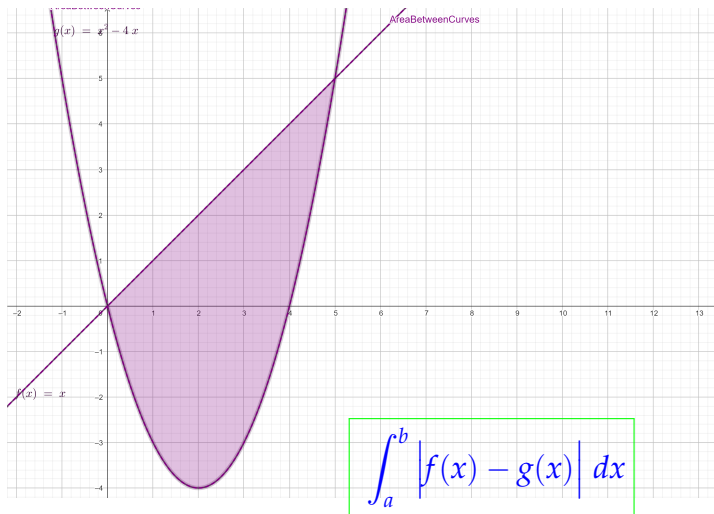
$$\text{Total Distance} = \int_a^b |v| dt$$

To calculate the distance the object travels during the time interval, we have to consider the intervals when $v(t) > 0$, while the particle moves to the right, and also the intervals when $v(t) < 0$, while the particle moves to the left. In both cases the **total distance** is computed by **integrating $|v|$, the speed**. Therefore,

$$\text{Total Distance traveled} = \int_0^2 |v(t)| dt = \int_0^2 |2 - 2t| dt$$

$$\begin{aligned} &= \int_0^1 (2 - 2t) dt + \int_1^2 (2t - 2) dt \\ &= (2t - t^2) \Big|_{t=0}^{t=1} + (t^2 - 2t) \Big|_{t=1}^{t=2} = 2(1) - 1^2 - 0 + 2^2 - 2(2) - (1^2 - 2) \\ &= 1 + 4 - 4 + 1 = 2 \end{aligned}$$

Area enclosed by two curves

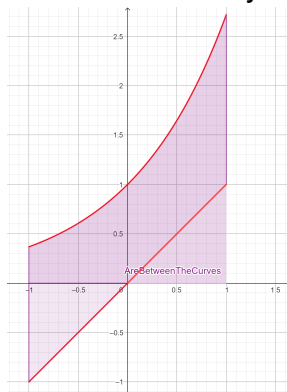


=area enclosed by the two curves $y = f(x)$ and $y = g(x)$

Area enclosed by two curves

Ex. 4. Find the area of the region bounded by the curves $y = e^x$, $y = x$ and the lines $x = -1$ and $x = 1$.

- Upper boundary curve is $y = e^x$
- Lower boundary curve is $y = x$



Area enclosed by two curves on $[-1, 1]$

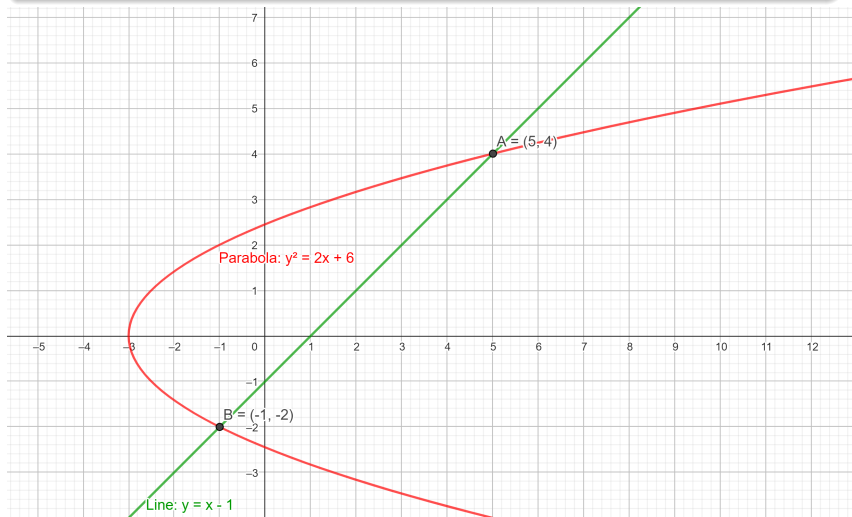
$$\text{Area} \int_{-1}^1 |f(x) - g(x)| dx$$

= area enclosed by the two curves $y = e^x$ and $y = x$

$$= \int_{-1}^1 [e^x - x] dx = \left[e^x - \frac{1}{2}x^2 \right] \Big|_{x=-1}^{x=1} = e - \frac{1}{2} - e^{-1} + \frac{1}{2} = e - e^{-1}$$

Area enclosed by two curves

Ex. 5. Find the area of enclosed by the parabola $y^2 = 2x + 6$ and the line $y = x - 1$.

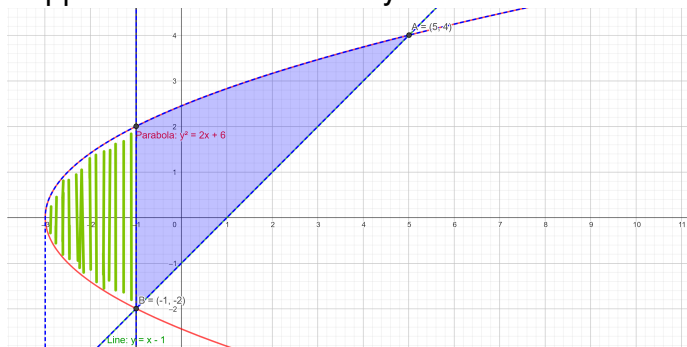


Area enclosed by the parabola $y^2 = 2x + 6$ and the line $y = x - 1$.

Solution. • Find the intersections between both curves:

$$2x+6 = y^2 = (x-1)^2 \Rightarrow 2x+6 = x^2-2x+1 \Rightarrow x = -1, x = 5$$

- Find upper and lower boundary curves on each interval



• On the left, $-3 \leq x \leq -1 \Rightarrow -\sqrt{2x+6} \leq y \leq \sqrt{2x+6}$

• On the right, $-1 \leq x \leq 5 \Rightarrow x-1 \leq y \leq \sqrt{2x+6}$

The bounded area is a union of 2 regions with different descriptions.

$$\text{Area} = A_1 + A_2$$

$$= \int_{-3}^{-1} \left[\sqrt{2x+6} - (-\sqrt{2x+6}) \right] dx + \int_{-1}^5 \left[\sqrt{2x+6} - (x-1) \right] dx$$

$$= \int_{-3}^{-1} \left[2(2x+6)^{1/2} \right] dx + \int_{-1}^5 \left[(2x+6)^{1/2} - x + 1 \right] dx$$

$$= \left[\frac{2}{3} (2x+6)^{3/2} \right]_{-3}^{-1} + \left[\frac{1}{3} (2x+6)^{3/2} - \frac{1}{2} x^2 + x \right]_{-1}^5$$

$$= \frac{16}{3} + \frac{56}{3} - 6 = \frac{54}{3} = 18 \quad \checkmark$$

More on Substitution Rule

$$\int f(g(x)) \cdot g'(x) dx = \int f(u) du$$

Example 6. Compute the integrals

a) $\int \frac{du}{\sqrt{1-u^2}}$

b) $\int_{-\sqrt{2}}^{-1} \frac{e^x}{\sqrt{1-e^{2x}}} dx$

Solution a) From the table of **antiderivatives**:

$$\int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1}(u) + C, \quad \int \frac{-1}{\sqrt{1-u^2}} du = \cos^{-1}(u) + C$$

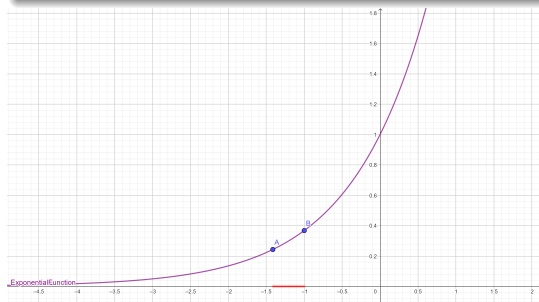
b) • Substitution $u = e^x \Rightarrow e^{2x} = u^2$ and $du = e^x dx$:

$$\int \frac{e^x}{\sqrt{1-e^{2x}}} dx = \int \frac{1}{\sqrt{1-u^2}} du = \sin^{-1}(u) + C = \sin^{-1}(e^x) + C$$

• Solve the definite integral:

$$\int_{-\sqrt{2}}^{-1} \frac{e^x}{\sqrt{1-e^{2x}}} dx = \left[\sin^{-1}(e^x) + C \right]_{-1}^{-\sqrt{2}} = \sin^{-1}(e^{-\sqrt{2}}) - \sin^{-1}(e^{-1})$$

Why is it possible to evaluate the definite integral?



- $[-\sqrt{2}, -1] \subset (-\infty, 0)$ then $0 < e^{-\sqrt{2}}, e^{-1} < 1$
- On the other hand, the inverse function

$$\sin t : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \rightarrow [-1, 1]$$

$$\sin^{-1} x : [-1, 1] \rightarrow \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$$

- Then it is possible to evaluate the definite integral:

$$[a, b] = [e^{-\sqrt{2}}, e^{-1}] \subset \text{Dom}(\sin^{-1}(x))$$